Complex Analysis Homework 7

1. Show that $g_k(x) = \dfrac{x^k}{1+x^k}$ converges pointwisely on $[0, \infty)$. What is the limit function? Show that $g_k(x)$ does not converges uniformly on any interval *I* which contains 1. Furthermore, show that $g_k(x)$ converges uniformly on $[r, \infty)$ for any $r > 1$.

2. (a) Verify that the identity $\mathrm{Log}\left(\dfrac{1+z}{1-z}\right) = \mathrm{Log}(1+z) - \mathrm{Log}(1-z)$ holds when $|z| < 1$. Then, using the Maclaurin expansions of $\mathrm{Log}(1 + z)$ and $\mathrm{Log}(1-z)$, find the Maclaurin expansion of $\mathrm{Log}\left(\dfrac{1+z}{1-z}\right)$.

(b) Find the power series expansion of the principal branch $\mathrm{Tan}^{-1}(z)$ of the inverse tangent function about $z = 0$. What is the radius of convergence of the series?

(c) Expand $\mathrm{Log}(1 + iz)$ and $\mathrm{Log}(1-iz)$ in power series about $z = 0$. By comparing power series expansions, establish the identity $\mathrm{Tan}^{-1}(z) = \dfrac{1}{2i}\mathrm{Log}\left(\dfrac{1+iz}{1-iz}\right)$ when $|z| < 1$.

3. Express the following function in power series at z_0 and find their radii of convergence.

(a) $\dfrac{e^z}{1+z}$ at 0

(b) $\dfrac{z^2-1}{z^4-1}$ at 2

4 (a) Show that $f(z) = \int_0^1 \dfrac{\sin zt}{t}\,dt$ is an entire function by obtaining a power series expansion for f.

(b) Show that $f'(z) = \int_0^1 \cos zt\,dt$ by using the power series for f.

5. Consider the following infinite series of complex functions $\sum_{k=1}^{\infty} \frac{\sin kz}{k^2}$. Show that it is absolutely convergent when z is real, but that when z is non-real, it becomes divergent.